GLOBAL GEOAI AGENT

A concept research paper for an autonomous, voice-controlled Earth intelligence platform integrating NASA satellite imagery, geospatial analytics, global event data and predictive AI.

Core vision

One interactive digital Earth that helps a person investigate what is happening across the planet, understand the evidence, explore historical changes and monitor emerging risks.

RESEARCH CONCEPT • SYSTEM DESIGN • IMPLEMENTATION ROADMAP



Prepared as a practical design blueprint. This document describes a proposed system, not a claim that a complete global monitoring system already exists.

Abstract

This paper proposes a Global GeoAI Agent: an autonomous, voice-controlled geospatial intelligence platform that combines satellite imagery, environmental observations, natural-event feeds, geographic information systems (GIS), specialist AI models and a conversational interface. Its central interface is an interactive 3D globe inspired by futuristic command-centre dashboards. Users can ask questions in natural language, request a geographic investigation, switch data layers, compare dates, and receive evidence-linked explanations and alerts.

The design places NASA's open Earth-observation ecosystem at its core. NASA Worldview supports interactive exploration of satellite imagery, while the Global Imagery Browse Services (GIBS) exposes imagery through standardized web services that developers can integrate into their own applications. NASA FIRMS can contribute active-fire detections. Other public and licensed providers can supply weather, seismic, ocean, transport, news and economic data. The system must preserve source attribution, observation time, uncertainty and coverage limits.

The proposed platform is not an omniscient system and does not provide a universal live video feed of Earth. Different instruments observe different phenomena at different times, spatial resolutions and revisit intervals. The goal is to create a reliable, extensible intelligence layer that continuously updates its view of selected data sources and investigates questions using verifiable evidence.

Keywords: GeoAI, agentic AI, Earth observation, satellite imagery, NASA GIBS, NASA Worldview, GIS, digital twin, event detection, environmental monitoring, voice interface.

1. Problem statement and motivation

Earth information is distributed across many specialized systems. Satellite images, weather forecasts, earthquake reports, fire detections, river measurements, news reports and historical datasets often use different formats, geographic references, timestamps and update schedules. A person researching an event may need to open multiple websites, interpret different maps and manually compare information. This creates friction, makes it difficult to see relationships between events, and can cause stale or conflicting information to be mistaken for current fact.

The proposed agent addresses this fragmentation by providing a unified geographic interface and an autonomous workflow. Rather than simply answering from a language model's stored knowledge, it should identify the user's intent, retrieve relevant source data, perform spatial and temporal analysis, check evidence quality, and explain its conclusions with links to the underlying data.

Primary research question: How can open Earth-observation data, geospatial computation and an agentic software architecture be combined into an interactive system that helps users investigate global conditions while clearly communicating data freshness, source reliability and uncertainty?

2. Goals and non-goals

- **Unified view:** Display selected global datasets on one interactive 2D/3D map or globe.
- **Satellite imagery:** Integrate NASA imagery layers and historical date controls.
- **Natural-language operation:** Let voice and text commands control the map and launch investigations.
- **Autonomous investigation:** Select tools, retrieve evidence, run analyses and synthesize results.
- **Proactive monitoring:** Check configured feeds and raise threshold-based alerts.
- **Traceability:** Show source, observation time, retrieval time, processing status and uncertainty.
- **Extensibility:** Add specialist feeds and models without rebuilding the whole platform.

Non-goals: the initial system should not claim access to every satellite, private sensor, military feed, individual's location or event on Earth. It should not treat generated explanations as evidence, promise guaranteed predictions, or automatically take high-impact actions based on uncertain analysis.

3. Data sources and satellite integration

NASA imagery should be a first-class data layer inside the application. The recommended strategy is to use NASA's public imagery services directly for visualization, and use downloadable scientific datasets when deeper analysis is required. These are related but distinct workflows: a visualization tile is optimized for display, while a scientific data product may contain measurements, quality flags and metadata needed for analysis.

Source / service	Role in the system	Important qualification
NASA Worldview	Explore global satellite imagery and historical dates; useful for validating layers and investigating events.	Imagery availability and delay depend on the product and instrument.
NASA GIBS	Serve imagery visualization layers through WMTS, WMS and tile endpoints for a custom map or globe.	Use the documented layer identifier, projection, date and attribution; not every layer has the same time coverage.
NASA FIRMS	Retrieve satellite-derived active-fire and thermal-anomaly detections for monitoring workflows.	A detection is not automatically a confirmed wildfire; false positives and detection limits must be considered.
NASA Earthdata Search	Discover and access underlying Earth science datasets for deeper processing and historical comparisons.	Some datasets require accounts, data preparation or substantial storage/compute.
NASA EONET	Use curated natural-event metadata to discover events and related imagery layers.	Event catalogues are not exhaustive ground truth for every event.
Other providers	Weather, seismic activity, river levels, ocean conditions, transport, public alerts, news and economic statistics.	Check licence, geographic coverage, rate limits, update interval and provider reliability.

Recommended NASA references

- NASA Worldview: https://worldview.earthdata.nasa.gov/
- NASA GIBS developer documentation: https://nasa-gibs.github.io/gibs-api-docs/
- NASA FIRMS: https://firms.modaps.eosdis.nasa.gov/
- NASA Earthdata Search: https://search.earthdata.nasa.gov/
- NASA EONET: https://eonet.gsfc.nasa.gov/

Implementation note: GIBS imagery is served as standardized visualizations. For direct integration, use the official API documentation and visualization product catalog to select supported layers, map projections and time dimensions. NASA requests acknowledgement when GIBS imagery is used in applications or presentations.

4. System architecture

The architecture separates the conversational agent from data connectors, geospatial processing, model execution and evidence storage. This separation makes the platform easier to test and allows new sources to be added without rewriting the interface.

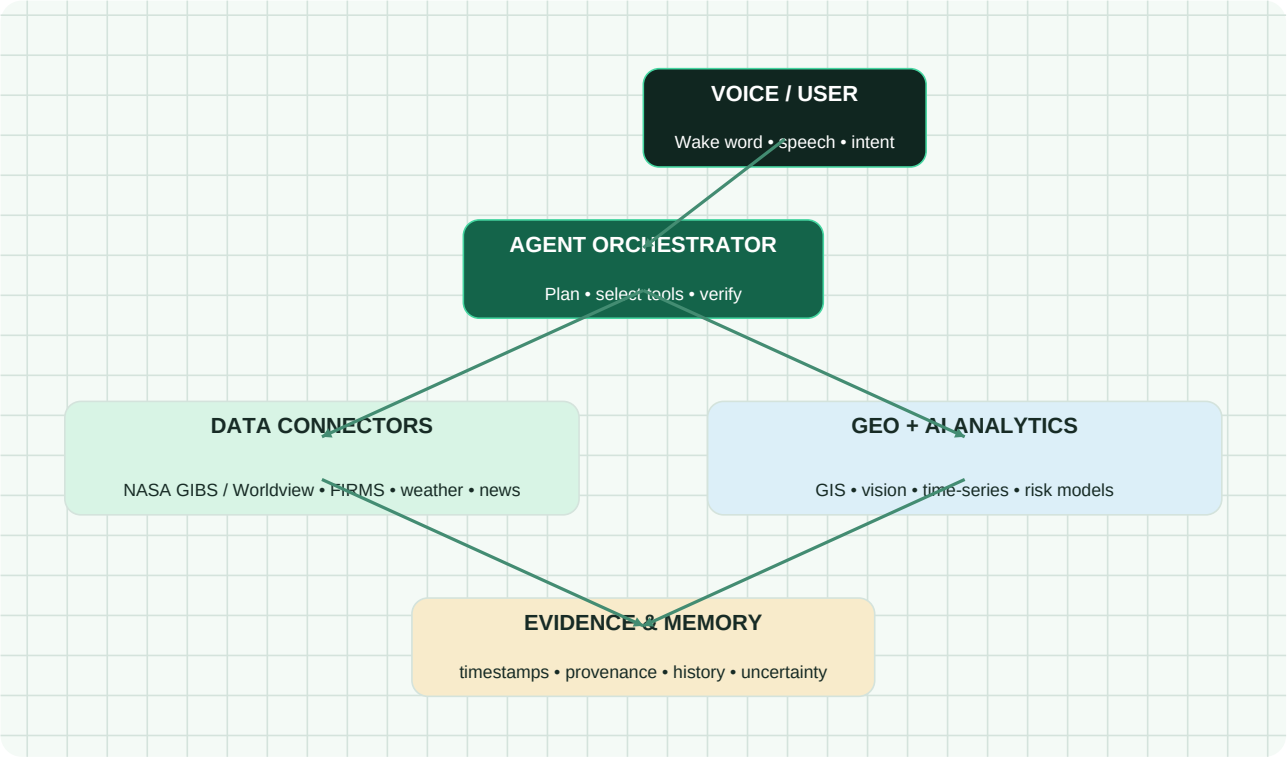


Figure 2. Proposed architecture. Arrows represent the main information flow; a production implementation should also support retries, error reporting, caching and feedback from validation to planning.

4.1 Core components

User interface. A persistent globe, searchable locations, layer toggles, time slider, alert panel, source details and voice controls.

Voice subsystem. Wake-word or push-to-talk detection, speech-to-text, command interpretation and text-to-speech. The wake-word listener should be configurable and provide a clear privacy indicator.

Agent orchestrator. Breaks a request into steps, selects tools, checks permissions, tracks progress, handles failures and assembles the final explanation.

Connector layer. Provider-specific adapters normalize APIs, data schemas, authentication, pagination, rate limits and retry behaviour.

Geospatial engine. Performs coordinate transforms, spatial joins, region filtering, raster/vector analysis, distance calculations and time-based comparisons.

Specialist models. Use computer vision, anomaly detection, time-series forecasting or domain-specific models only where validated data and evaluation are available.

Evidence store. Persists event records, source URLs, raw/normalized timestamps, processing history, user-defined watch areas and audit logs.

4.2 Suggested technology stack

Layer	Practical starting choice
Frontend	HTML/CSS/JavaScript; Three.js or CesiumJS for a 3D globe; a chart library for trends.

Layer	Practical starting choice
Backend	Python + FastAPI; REST endpoints for commands and metadata; WebSockets or server-sent events for updates.
Spatial storage	PostgreSQL + PostGIS for spatial queries; object storage for larger imagery/data files.
Background tasks	A scheduler first; add a task queue and message broker when feed volume grows.
Analysis	Python geospatial and scientific libraries; model-specific runtimes only when required.
Monitoring	Structured logs, health checks, source freshness metrics, error counters and audit trails.

5. Agentic behaviour and example workflow

An agentic system does more than answer one prompt. It plans a sequence of actions, uses tools, observes the results, and adjusts its next step. For this platform, autonomous behaviour should be bounded by explicit policies: the agent may retrieve public data and update the display automatically, but actions with safety, privacy, financial or operational consequences should require appropriate human approval.

Example voice command

“Show me the latest available NASA satellite imagery of eastern India, display active-fire detections, compare it with imagery from seven days ago, and tell me what changed.”

1. Parse intent. Identify the requested region, imagery layer, comparison dates and desired spoken summary.

2. Check data availability. Find supported imagery products and dates; show a clear message if a layer or date is unavailable.

3. Retrieve evidence. Load GIBS imagery and query a fire-detection feed for the selected geographic bounds and time window.

4. Align and validate. Normalize timestamps and coordinates, remove duplicate records, and distinguish observation dates from retrieval times.

5. Analyze. Compare available imagery and event detections. If the comparison requires scientific image processing, use suitable source data rather than relying only on display tiles.

6. Update the globe. Apply selected layers, mark event locations and display a historical timeline.

7. Explain with provenance. Speak a concise result and show source links, dates, limitations and uncertainty beside the map.

6. Priority use cases

Use case	What the agent does	Boundary / caution
Disaster awareness	Combines official alerts, event feeds, satellite observations and exposed-population layers.	Not a replacement for emergency authorities or official warnings.
Cyclone monitoring	Shows observed storm imagery, official track forecasts and affected coastal regions.	Forecast tracks change; clearly label forecast time and issuing agency.
Wildfire analysis	Maps active-fire detections, smoke imagery and nearby communities where data is available.	Thermal detections do not always equal confirmed fires or define fire boundaries.
Environmental change	Compares imagery and indicators across dates to investigate vegetation, water or land-cover changes.	Season, cloud cover, sensor differences and processing affect comparisons.
Regional briefing	Summarizes verified changes for a chosen area over a specified period.	Report data coverage and unresolved contradictions.
Education / research	Lets learners explore Earth systems, historical imagery and evidence-backed geographic questions.	Cite datasets and explain scientific uncertainty.

7. Interface design principles

The existing dark command-centre style can remain intact. New capability should be added behind the existing visual structure rather than forcing a complete redesign. The globe is the primary navigation surface; side panels should reveal context only when a layer, event or region is selected.

Central globe. Pan, rotate, zoom, select a region, inspect an event and switch between globe and flat-map modes.

Layer manager. Group layers by weather, satellite imagery, natural events, environment, transport and socio-economic indicators.

Timeline. Select date/time; compare two periods; show data gaps and each layer's latest available observation.

Evidence drawer. Display source, provider, product name, timestamp, retrieval time, quality notes and link to original data.

Voice status. Show when the wake-word listener is active, when audio is being processed, and when speech output is playing.

Alert rail. Prioritize alerts by severity and confidence, with mute, acknowledge and review controls.

Data freshness badges. Use explicit labels such as LIVE STREAM, RECENT OBSERVATION, HISTORICAL, FORECAST, STALE or UNAVAILABLE.

Design rule: a glowing animation, connection line or synthetic marker must never be presented as a real satellite track, live sensor, or measured value unless it is backed by a verified data record.

8. Safety, privacy and reliability

- **Provenance:** Store the original source and timestamp for every important claim or map feature.
- **Freshness:** Display acquisition/observation time separately from the time the platform retrieved the record.
- **Uncertainty:** Separate measured observations, reports, model estimates and forecasts visually and in spoken explanations.
- **Human oversight:** Critical alerts should link to official authorities and avoid claiming that the AI is an emergency service.
- **Privacy:** Process wake-word audio locally where feasible, expose microphone status, minimize retention and never record continuously without clear consent.
- **Security:** Keep API credentials server-side, use least-privilege access, validate tool arguments and audit sensitive actions.
- **Abuse resistance:** Restrict attempts to infer private individual movements or combine datasets to target people.
- **Resilience:** Handle unavailable providers, API limits, stale feeds, malformed records and network failures without silently fabricating data.

9. Evaluation plan

Evaluation should measure reliability and usefulness, not visual sophistication alone. Recommended metrics include:

Dimension	Example metric	Targeting approach
Source reliability	Percentage of displayed records with valid source, timestamp and geographic reference.	Aim for 100% provenance on source-derived records.
Freshness	Age of the latest record per feed and percentage of stale-layer warnings shown correctly.	Set per-provider thresholds; do not use one universal threshold.

Dimension	Example metric	Targeting approach
Agent correctness	Tool selection success, geographic scope accuracy, and supported claims in generated summaries.	Evaluate against curated test questions and expected source records.
Forecast quality	Domain-specific error scores and calibration on held-out historical data.	Require baseline comparisons and independent validation before alerts.
Usability	Time to find an event, voice-command success rate and user task completion.	Test with representative users and real workflows.
Resilience	Recovery from timeouts, API rate limits and malformed responses.	Use simulated outages and provider failure tests.

10. Development roadmap

Phase 1 — Reliable globe

Build the map/globe, location search, layer controls and a basic evidence panel. Integrate one NASA imagery layer through GIBS and verify date handling.

Phase 2 — First real feeds

Add NASA FIRMS and one natural-event feed. Implement a normalized event schema, caching, retries and visible data-freshness indicators.

Phase 3 — Voice control

Add speech recognition, text-to-speech, wake-word or push-to-talk operation, and commands for navigation, layer switching and date selection.

Phase 4 — Agent planner

Create a tool registry and planner that chooses connectors and analysis steps. Require source-linked evidence in final answers.

Phase 5 — Historical analysis

Store normalized event history, add time-series charts and compare imagery or measurements across selected periods.

Phase 6 — Specialist intelligence

Introduce validated models for selected tasks such as anomaly detection or environmental change; compare performance with simple baselines.

Phase 7 — Scale and harden

Add monitoring, permissions, user-configurable alerts, deployment automation, cost controls, security reviews and operational documentation.

11. Main engineering challenges and mitigation

Inconsistent data formats	Build connector adapters and a shared schema with standardized UTC timestamps, coordinates, units and source identifiers.
Imagery latency and coverage	Expose product-specific update times, support alternate layers and avoid calling delayed imagery “live video.”
False event interpretations	Use source-specific quality checks, corroboration and explicit labels for detections versus confirmed events.
High compute and storage cost	Start with display tiles and metadata; download large source datasets only for justified analyses; cache repeated requests.
LLM hallucinations	Ground summaries in retrieved records, require citations/source links, validate tool outputs and say when evidence is insufficient.
Over-automation	Limit autonomous actions to low-risk operations; require approval for consequential actions and provide a clear audit trail.

12. Conclusion

A Global GeoAI Agent is a feasible long-term software architecture if it is built incrementally and grounded in real data. NASA Worldview and GIBS provide a practical starting point for satellite imagery integration; NASA FIRMS and event catalogues can add selected event signals; and a geospatial backend can align these sources for analysis. An agentic layer can then interpret voice commands, choose tools, compare evidence, operate the globe and prepare proactive briefings.

The central challenge is not to draw a convincing futuristic interface. It is to maintain an accurate, time-aware and auditable representation of the data that the interface displays. The strongest version of this system will be explicit about what it knows, how it knows it, when the data was observed, what remains uncertain and which source should be consulted for decisions.

Recommended first milestone: a working globe that displays a real NASA GIBS imagery layer, allows date selection, overlays one real event feed, and responds to a voice command by changing the map. This small but verifiable milestone establishes the foundation for the larger autonomous Earth-intelligence vision.

References and official resources

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[7] NASA Open APIs portal. https://api.nasa.gov/

[8] NASA GIBS, “Accessing via GIS Applications.” https://nasa-gibs.github.io/gibs-api-docs/gis-usage/

[9] NASA GIBS / Worldview source repository. https://github.com/nasa-gibs/worldview

Data-use and attribution note

Before deployment, review the current terms, attribution requirements, authentication requirements, rate limits and availability for every provider and dataset. NASA GIBS documentation asks users to acknowledge imagery provided by NASA’s Global Imagery Browse Services (GIBS), part of NASA’s Earth Science Data and Information System (ESDIS). Other datasets may have separate attribution or licensing conditions.

Scope note

This is a concept research and engineering plan. It does not report experimental results, prove predictive accuracy or imply official NASA partnership. All predictions, alerts and summaries described are proposed capabilities that require implementation and validation.